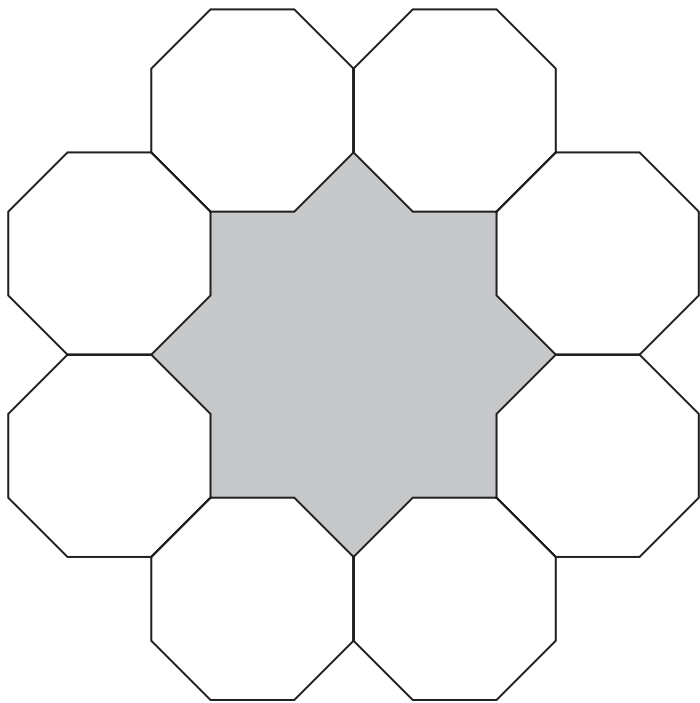


24 The diagram shows 8 identical regular octagons joined to enclose a shaded shape.



Each octagon has sides of length a .

Find, in terms of a , an expression for the area of the shaded shape.

Give your answer in the form $p(2 + \sqrt{2})a^2$ where p is an integer.
You must show all your working.

(Total for Question 24 is 5 marks)